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Stable Isotope Resolved Metabolomics Classification of Prostate Cancer Cells using Hyperpolarized NMR Data

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Supporting information is available.

Abstract: Metabolic fingerprinting is a strong tool for characterization of biological phenotypes. Classification with machine learning is a critical component in the discrimination of molecular determinants. Cellular activity can be traced using stable isotope labelling of metabolites from which information on cellular pathways may be obtained. Nuclear magnetic resonance (NMR) spectroscopy is, due to its ability to trace labelling in specific atom positions, a method of choice for such metabolic activity measurements. In this study, we used hyperpolarization in the form of dissolution Dynamic Nuclear Polarization (dDNP) NMR to measure signal enhanced isotope labelled metabolites reporting on pathway activity from four different prostate cancer cell lines. The spectra have a high signal-to-noise, with less than 30 signals reporting on 10 metabolic reactions. This allows easy extraction and straightforward interpretation of spectral data. Four metabolite signals selected using a Random Forest algorithm allowed a classification with Support Vector Machines between aggressive and indolent cancer cells with 96.9% accuracy, -corresponding to 31 out of 32 samples. This demonstrates that the information contained in the few features measured with dDNP NMR, is sufficient and robust for performing binary classification based on the metabolic activity of cultured prostate cancer cells.

Introduction

Metabolomics, the profiling and study of the small molecular products of a system, is a powerful tool for gaining insight into the biochemical state of an organism.[1] The snapshot of metabolites at a certain time point - the so-called metabolic fingerprint - can be correlated to the phenotype of the organism and can therefore be used for diagnostic purposes.[2] Metabolic fingerprints were, for instance, used as a basis for classifying the aggressiveness of cancers.[3] Machine Learning (ML) algorithms are especially useful for the classification of metabolic fingerprints as belonging to distinct biologies.[4,5] Such algorithms can be applied to make a feature based separation as well as to identify which measured features are important for a successful classification.[5,6]

Nuclear Magnetic Resonance (NMR) and Mass Spectroscopy (MS) are the two most popular analytical techniques for generating metabolic fingerprints. NMR is reproducible and quantitative within a large dynamic range without need for authentic standards and it provides specificity on an atomic level. It is however relative to MS insensitive.[7] Metabolic fingerprints obtained with NMR are most often acquired as a single sampling of metabolites belonging to the investigated biological system. Biofluids or extracts from cells or tissue are directly processed and metabolite levels are quantified and compared. Such analysis provides a snapshot into a given biological system. Alternatively, biological function may be measured by profiling the metabolic activity in a network of biochemical transformations. In this case metabolic transformations may be traced by stable isotopes, e.g. ^{13}C . [8] Such method has been called stable isotope resolved metabolomics (SIRM). [9,10] Metabolite levels and pathway activities represents complementary information about the biological system and both are needed for metabolic understanding. [11]

Due to its high sensitivity ^1H NMR is widely used in metabolomics. These NMR experiments are however challenging to use for bio-marker identification due to extensive signal overlap that hampers both metabolite identification and quantification. Issues that are even more pertinent when traced metabolites are used for analysis of pathway activity due to further signal overlap from coupling satellites. In such studies NMR methods that take advantage of the larger chemical shift ranges of low-gamma nuclear spins are advantageous. These experiments, however, come with the penalty of longer acquisition times with sample stability as a complication.

A sensitivity boost to NMR may be obtained by applying dissolution Dynamic Nuclear Polarization (dDNP), where the polarization of the nuclei is enhanced *ex situ* before the NMR spectra are recorded. [12] Using dDNP, the NMR signal

can be enhanced several orders of magnitude compared to room temperature equilibrium. We have recently developed a method combining dDNP with isotope labelled metabolomics NMR. Specifically, we showed that metabolic activity can be measured reproducibly and quantitatively using [U- $^{13}\text{C}_6$, d $_7$]D-glucose as a tracer for glycolytic activity in cancer cells from dDNP signal enhanced 1D- ^{13}C -NMR spectra of transformation products.[13] dDNP enhanced ^{13}C -NMR data has been shown to meet the repeatability demands of metabolomics, which further supports the methods potential as a complementary analytical tool.[14] The dDNP 1D- ^{13}C -NMR spectra are sparse and metabolite signals are readily identified and quantified.

Machine learning (ML) is by now an integrated tool in metabolomics where it is used e.g. for prediction of cancer progression from discriminations based on metabolite levels.[15] ML is, however, not commonly used on isotope labelled metabolite data reporting on metabolic transformations. Generally, NMR data from studies involving tracing of isotope labelled metabolites are sparse compared to those involving measurement of metabolite levels, which should be taken into account when choosing an appropriate ML algorithm: If an input data matrix (consisting of signal integrals in a number of samples) contains more NMR signals than samples this renders a proper validation of the separation and feature selection important for classification.[5] ML algorithms which are commonly used in such situations often perform worse in terms of prediction accuracy than other classifiers when challenged with lower number of variables.[16]

The present study is the first study to evaluate whether dDNP ^{13}C -NMR may be used to provide fingerprints of metabolic activity useful for classification of cancer cell phenotypes. The study goal was three-fold: To investigate whether it was possible to make a robust multi-class classification based on few NMR signals and capture differences between four prostate cell lines. Secondly, to evaluate if the use of metabolic activity measurements such as those obtained with dDNP ^{13}C -NMR can be used to separate aggressive cancer cell lines from indolent cancer cells in a binary classification. And thirdly through a feature analysis to examine the importance ranking of NMR signals for identified metabolites in the binary separation.

Results

dDNP ^{13}C -NMR spectra were obtained from four different prostate cancer cell lines: Two highly invasive, androgen independent cell types: PC3 (metastatic, derived from bone marrow) and DU145 (metastatic derived from brain), and two low invasive, androgen dependent cell types: LNCaP (early metastasis derived from lymph node), and PNT1A (PNT, immortalized cells from healthy prostate). A total of 32 samples (8 from each cell type), were made by the method described in Lerche et al.[13] In brief, cell samples were incubated with [U- $^{13}\text{C}_6$, d $_7$]D-glucose for 30 min, before metabolism was quenched and metabolites were extracted with perchloric acid. The samples were lyophilized, and the metabolic mixture was prepared for dDNP in a mixture of water, glycerol and trityl radical with addition of a reference compound (HP001, 1,1-bis (hydroxymethyl) cyclopropane) for referencing and quantification. All samples were hyperpolarized with DNP for 1.5 hours, dissolved in phosphate buffer (pH 7.4) and directly transferred to a 9.4 T NMR magnet for recording of 1D ^{13}C -NMR spectra (see suppl. Inform. for detailed methods). The carbon NMR signals were straightforwardly identified as belonging to specific atoms in energy/glycolytic metabolites by consulting available databases.[17] This was done using distinct chemical shifts of carbonyl carbons and by comparing ^{13}C - ^{13}C coupling constant pattern with simulations using the NMR analysis tool MNova,[18] since all metabolite signals are derived from uniformly ^{13}C -labelled glucose, Figure 1. In comparison to conventional SIRM NMR using 1D ^1H -NMR with ^{13}C -couplings for analysis the dDNP 1D ^{13}C -NMR spectra has less overlap, SI Figure S2. Whereas ten metabolites were directly identified and quantified in the dDNP 1D ^{13}C -NMR spectra only two of these, alanine and lactate, could be unambiguously identified and quantified from the 1D ^1H -NMR spectra.

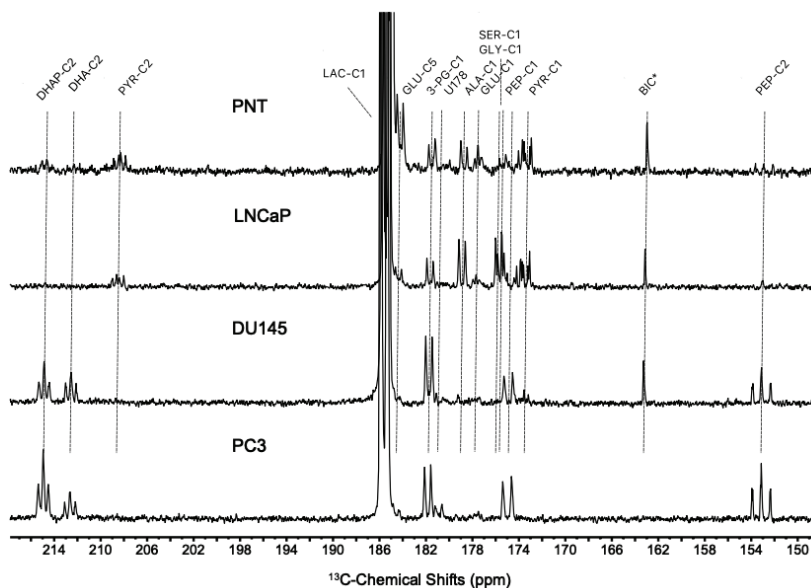


Figure 1 Carbonyl region of dDNP ^{13}C -NMR metabolic fingerprint. One representative spectrum of metabolite extracts from each of the four prostate cell types (PNT, LNCaP, DU145 and PC3) are shown. Full spectra are shown in SI Figure S1. Identification of the signals as belonging to specific carbon atoms are indicated above the top spectrum, see Table 1 for name identification. * indicates that bicarbonate (BIC) was not reliable as a labelled metabolite in the analysis due to the applied extraction method and thus not included in the analysis.

The spectral features used for further analysis are listed in Table 1. Signals representative of identified metabolites was included in the ML analysis as quantified peak areas relative to the internal standard. One carbon signal representative of each identified metabolite was included in the analysis; in general long T_1 carbons (e.g. carbonyl atoms) and carbon signals with less overlap in the spectra were prioritized as representatives, as an example of the latter PEP-C2, which lies completely isolated in the spectra was included rather than PEP-C1.

Table 1 List of spectral features for which integrals were used as input in the ML algorithm. Chemical shifts were referenced to an added internal standard compound HP001 at 25.7 ppm. The abbreviations reflect the carbon position. Due to overlap the signals from amino acids, serine and glycine were integrated together.

Chemical shift (ppm)	Assigned compound	Abbreviation
214.6	Dihydroxyacetone phosphate	DHAP-C2
185.3	Lactate	LAC-C1
184.25	Glutamate	GLU-C5
181.5	3-Phosphoglycerate	3PG-C1
180.6	Unidentified amino-acid	U178
178.8	Alanine	ALA-C1
175.5	Serine and Glycine	SER-GLY-C1
173.3	Pyruvate	PYR-C1
152.9	Phosphoenolpyruvate	PEP-C2
92.1	Glyceraldehyde 3-phosphate	G3P-C1

On average, there were visible differences in the metabolite production between cell lines. An analysis of variance (ANOVA) test on the distribution of all measured integrals over 8 samples for every cell type showed that there was a significant difference between the four cell lines in the measured integrals of DHAP-C2, ALA-C1 and PEP-C2. Although the data precision was generally low the ANOVA test suggests that these metabolites may carry discrimination value in a classification between the four prostate cancer cell types. Also, pairwise statistics were performed between each cell line, for every metabolite with a T-test. This test showed significant difference for the metabolite DHAP-C2 between the aggressive cell type DU145 and the indolent PNT and LNCaP, respectively (Bonferroni corrected significance, $P < 5\%$), SI, Table S2.

Our first point of inquiry was whether a classification, using an appropriate ML algorithm, could be made to identify each sample to the cell line it came from. Contrary to conventional NMR, dDNP NMR spectra, using ^{13}C labelled glucose as tracer contain fewer spectral features. Bearing this in mind, it was important to carefully choose the classification tools in order not to overfit when the number of samples was similar to the number of features. Partial Least Squares-Discriminant Analysis (PLS-DA) is widely used for classification with NMR in metabolomics,[4,19] but have been shown to under-perform and over-fit, compared to other methods, especially in situations with a low number of features.[16,20-22] To perform this multiclass classification on the dDNP ^{13}C -NMR data we chose instead Principal

Component-Discrimination Function Analysis (PC-DFA). PC-DFA has been applied in a wide variety of metabolomics studies and have been shown to perform robustly with a small number of features.[5,16,20] Since PC-DFA is a parametric function and thus performs optimally on normal distributed data, a box-cox transformation[23] was applied to the data-matrix. In the first step of the PC-DFA, the Principal Component Analysis (PCA), 7 principal components were needed to account for 95.68% of the variance in the original data. These components were applied in the second step of the algorithm where a supervised Linear Discriminant Analysis (LDA) was used to construct a set of discriminant functions, which optimizes the distance between the cell lines and minimizes the distance between samples of the same cell line. Leave-One-Out cross validation were applied to the LDA. The results of the PC-DFA analysis are shown in Figure 2. The unsupervised principal component analysis represented in Figure 2A by the first two PC of the linear transformation of the scaled data is clearly not sufficient to separate the metabolic fingerprints by cell line. There is a tendency for the PNT and LNCaP samples to collect in the upper left of the plot, the DU145 samples to be near the middle and PC3 to generally be to the lower right. With supervision this improved and all data points (8 samples for each of the 4 cell lines) confined within two standard deviation point clouds of the respective cell lines, except two from the PNT cell line and one from the PC3 cell line, Figure 2B. The success rate measured as accuracy (the sum of true positives and true negatives divided with the total number of observations) was for this model with leave-one-out cross validation the success rate of prediction from this model was 78.125% (25 out 32 samples). The encountered prediction errors were between cell lines characterized by being aggressive (PC3 and DU145) or between those characterized as indolent (PNT and LNCaP) respectively, Table 2.

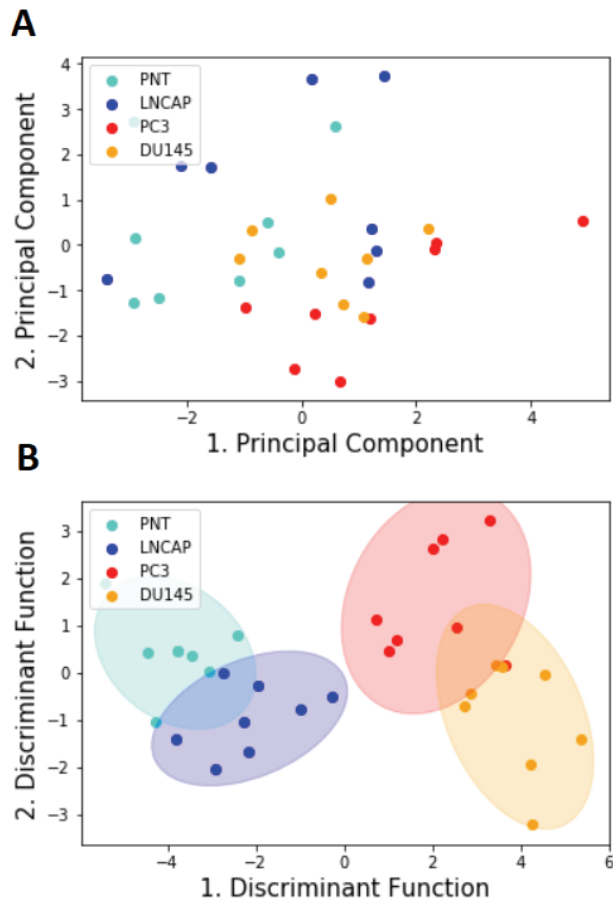


Figure 2 A) Illustration of PCA on box-cox transformed data. First PC contains 34.47% of the variance in the data and second PC 26.65%. A total of 7 components were needed to explain more than 95% of all data variance. B) Plot of the two first PC-DFA functions, illustrating the results of separation between cell lines. The data has been Box-Cox transformed and normalized, before PC-DFA was performed, with 7 features used from PCA capturing 95.68% variance of the original data. Colored eclipses represent 2 standard deviations of point clouds. PC3 (red) and DU145 (orange) are aggressive prostate cancer cell lines and PNT (cyan) and LNCaP (blue) are characterized as indolent.

Table 2 Error matrix for PC-DFA classification, with Leave-One-Out cross validated predictions.

Predicted:	PNT	LNCaP	PC3	DU145
Actual:				
PNT	6	2	0	0
LNCaP	3	5	0	0
PC3	0	0	7	1
DU145	0	0	1	7

These results prompted our second point of enquiry using the dDNP ^{13}C -NMR data to discriminate cell lines by their aggressiveness. The PC-DFA algorithm was merely instructed to separate individual cell lines and not to separate for cell line aggressiveness, so the fact that the prediction errors were made only between cell lines of similar aggressiveness suggested that a binary classification of the data may be more robust. An accurate discrimination between aggressive and indolent cancer at the time of diagnosis is a pressing clinical need in managing prostate

cancer.[24] Taken together this encouraged us to make a binary classification with the aim to separate between the highly invasive (DU145 and PC3, grouped as aggressive) and low invasive (PNT and LNCaP, grouped as indolent) cell lines. The LNCaP cell line is an early metastasis and as such not per se seen as a non-aggressive cell line. It is however androgen dependent and show a low invasiveness.[25] Since samples from this cell line separated with PNT in the multiclass separation it was therefore sorted together with PNT samples as an indolent cell line. A binary separation allows the use of Support Vector Machines (SVM) as a classifier. SVM has been shown to be particularly robust towards overfitting, compared with other classification methods commonly used in metabolomics.[5] This algorithm is non-parametric and thus enable data analysis without the need for transformations other than mean centering and scaling.[5,22] With the SVM algorithm a plane is found in the feature-space to maximize the separation between samples belonging to the two classes. Prior to the data analysis with SVM it is useful to make a dimensionality reduction, this has been shown to be particularly beneficial for small training datasets.[26] The dimensionality reduction was performed by the Random Forest (RF) algorithm. This algorithm provides at the same time a means to perform feature ranking. RF uses multiple decision trees, each made from a subset of features and samples, using the remaining samples for internal validation. The feature importance is measured directly by judging the importance of each individual feature in building the trees to separate the classes. By comparing the different forests to data with shuffled classes, a feature ranking is obtained which is unbiased to the covariance of the features.

The feature ranking made from classifications of the dDNP ^{13}C -NMR data with RF (see SI for details on method) is shown in Figure 3.

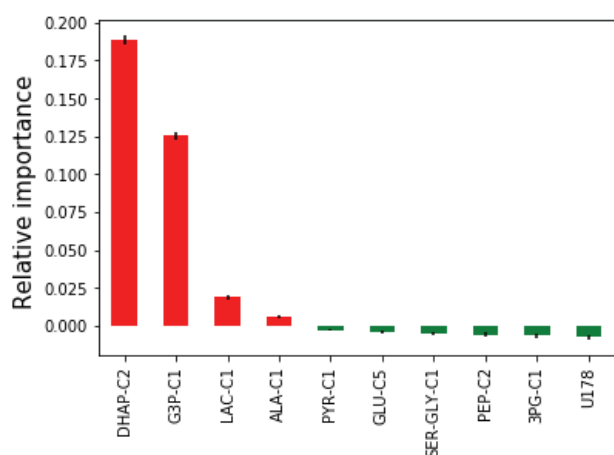


Figure 3 Feature importance measured as the change in predictive power of Random Forest (RF) classification when compared to randomized values in testing data sets. A positive score (in red) means the classification became worse when the testing values were shuffled. Black bars are the standard error.

From this feature ranking it can be seen that only signals representing DHAP, G3P, lactate and alanine contribute positively to the classification of aggressive vs. indolent prostate cancer samples. Data from the remaining metabolites are either unnecessary or creating noise in this algorithm. It is important to note here, that the feature selection of RF is not perfect, especially when variables differ in scale,[27] such as the lactate signal did in this study. However, the ranking is useful when as here it is used for constructing feature sets for classification with the SVM algorithm. Following this feature ranking using the RF algorithm an SVM classification was performed, based on the features with high importance selected from the feature importance analysis.

Prediction successes from SVM are shown in Table 3. With leave-one-out cross validation a classification using all features, accurately predicted the aggressiveness of 93.75% (30 out of 32) samples. When including only the four features found to have positive impact on the RF predictive power, DHAP, G3P, lactate and alanine, the highest observed success rate, 96.88% (31 out of 32) samples, was achieved. This high prediction success rate was kept eliminating DHAP but keeping G3P, Lactate and alanine. Any other combination of three features resulted in a success-rate level identical to the test with all features included. If only one feature was included the best prediction was found using DHAP (90.63% success rate) whereas alanine alone performed the worst with a 62.5% success rate. A similarly low prediction success of 75% was obtained when lactate was used as the only feature.

Table 3 Leave-One-Out cross validating (n = 32) results of classification with SVM for different sets of metabolite integrals. The tested sets of metabolite integrals were chosen on the basis of a RF feature ranking.

Features included	Success rate
all	93.75%
DHAP, G3P, LAC-C1, ALA-C1	96.88%
DHAP, G3P, LAC-C1	93.75%
DHAP, G3P, ALA-C1	93.75%
DHAP, LAC-C1, ALA-C1	93.75%
G3P, LAC-C1, ALA-C1	96.88%
DHAP, G3P	90.63%
DHAP, LAC-C1	90.63%
DHAP, ALA-C1	90.63%
G3P, LAC-C1	81.25%
G3P, ALA-C1	93.75%
ALA-C1, LAC-C1	81.25%
DHAP	90.63%
G3P	84.37%
LAC-C1	75.00%
ALA-C1	62.50%

The SVM analysis showed that information on as few as three or four metabolites, G3P, lactate and alanine or these metabolites plus DHAP were needed for highest observed prediction value among the tested data sets. It could be expected that lactate would be a decisive metabolite based on the numerous studies suggesting diagnostic value of this metabolite for detection of prostate cancer malignancy both in animal studies[28,29] and in clinical studies.[30] However, the SVM analysis was significantly worse including only lactate in the analysis suggesting that accumulation of DHAP and G3P and lower alanine production adds crucial diagnostic information about the metabolic strategy of aggressive prostate cancer cells. Accumulation of the trioses, DHAP and G3P may provide evidence for a flexible glycolytic strategy of highly proliferating prostate cancer cells. This notion is supported by recent reports on strict regulation of ATP production by the presence of a less active isoform of pyruvate kinase (PKM2) to keep an increased glycolytic activity in highly proliferating cells.[31] Also accumulation of DHAP suggest a shift in glucose catabolism to glycerol, which fit well with the ability of late stage prostate cancer cells to be producers of de novo lipids independent on androgen regulation.[32]

Although it is early days for the dDNP ^{13}C -NMR metabolomics method it is promising as a complementary method to conventional SIRM NMR. The dDNP ^{13}C -NMR spectra allowed easy identification and quantification of the trioses important for the metabolomic classification. These metabolites would not have been identified in a conventional SIRM 1D ^1H NMR spectrum due to spectral overlap, (SI, Figure S2). NMR methods that take advantage of the larger chemical shift ranges of low-gamma nuclear spins could solve some of these issues with spectral overlap. An experiment such as high resolution ^1H - ^{13}C -HSQC where ^{13}C - ^{13}C -coupling information is available could prove to be an alternative valuable method.[8] These experiments crave however long acquisition times and thus puts demand on sample stability. While promising ultra-fast NMR acquisition methods with detection on ^1H are developing for metabolomics,[33] schemes for correspondingly fast ^1H - ^{13}C -HSQC are still in their early stage.[34] For the dDNP ^{13}C -NMR metabolomics method to make its way into broader application in the field of metabolomics it will in the future be important to make a dedicated comparative study to conventional metabolomics tools such as 1D ^1H NMR and 2D ^1H - ^{13}C -HSQC to understand the comparative benefits and challenges of each method.

This study showed that carefully chosen ML methods robust against overfitting and tailored the classification problem allowed both a semi successful multi-class separation and a highly successful binary discrimination of prostate cancer cell types. In particularly it was shown that metabolic fingerprints obtained with dDNP ^{13}C -NMR provided sufficient information to successfully classify 31 of 32 prostate cancer cell line samples as belonging to the aggressive or indolent cell type. Subsets of only three or four of the measured features (assigned to (DHAP), G3P, lactate and alanine) were sufficient to support this classification. A caveat to note with the design of this study, is however, that the assumption of independence between each sample is not met, since they are taken from the same four cell lines, and not e.g. individual animals. The method should therefore be further evaluated in a study involving relevant tissue samples. On such basis

this method is expected to meet requests in the field of metabolic imaging[35] and provide identity of metabolic activities important for selection of agent(s) central to specific diseases.

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Competing interests

The authors have no competing interests to declare.

Keywords: dissolution dynamic nuclear polarization • stable isotope resolved metabolomics • random forest • support vector machine • nuclear magnetic resonance

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